

Literature Review: Robotic Arc Welding with the UR5e

1. Background

Welding means joining two metal parts together by melting them along a line called a seam. A robot is good at this because it can move at a steady speed and keep the same angle the whole way, which is hard for a person to do for a long time. Right now, a lot of welding robots are simply "taught" a path once and then repeat it forever, which breaks down if the part moves or changes. Newer collaborative robots, or "cobots," are built to work safely next to people instead of behind a cage, so factories can use them for welding while a human still checks the quality of the weld.

2. Cobot Welding in the Real World and in Research

In industry: Universal Robots (the company that makes the UR5e) has been selling welding kits built around its arms since around 2017, and by 2021 they said welding was one of their fastest-growing uses. Companies sell full welding kits that attach a torch and a cart to a UR5e or UR10e arm, ready to use.

In research: Some researchers have added a 3D scanner or a camera next to the torch so the robot can find the seam by itself instead of needing it to be placed in an exact spot. The robot then builds a path for the torch to follow, which includes both where the torch is and which way it is pointing. That is the same basic problem this course asks us to solve: work out a path and make the arm follow it.

3. About the UR5e

The UR5e is a robot arm with six joints. It can carry up to 5 kg, which limits how heavy the welding torch and any cables or camera can be. It can reach about 850 mm from its base, which sets how big the work area can be. It repeats the same position to within 0.03 mm, which is important for following a seam accurately. It also has a sensor built into its wrist that can feel force, which is useful if we want the robot to react to touching something. Because it has six joints, it can point the torch in any direction we need, not just move it in a straight line.

4. Why We Picked the UR5e for Welding

- **It matches the assignment.** The project needs an arm with 4 to 7 joints, and the UR5e has 6. Welding needs the torch to both reach a point and point in a certain direction, and a 6-joint arm can do both at once.
- **It is a real industrial robot.** Companies actually use UR arms for welding today, so we can back up our choice with real examples instead of a made-up scenario.
- **It works with our tools.** A ready-made UR5e model is already available for MuJoCo, and UR arms are well supported by ROS 2, which is what we are using for this project.
- **It is the right size.** Our test parts are small, so the UR5e's reach and weight limit are enough. The bigger UR10e would be more than we need.

5. How This Connects to the Rest of the Project

Welding gives us a clear task to build the rest of the project around. We will work out the arm's geometry and use it to point the torch at the seam (Milestone 2). We will work out how fast each joint needs to move so the torch travels at a steady speed along the seam (Milestone 3). Then we will plan the full path, straight lines as well as curves and corners, and get the simulated robot to follow it with a controller (Milestone 4).